MILITARY SPECIFICATION

PROPELLANT M1
FOR USE IN CHARGE, PROPELLING, XM211

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers M1 Propellant for use in the XM211 propelling charge for the M198 & M109A2/A3 Howitzer.

1.2 Classification. The propellant shall be of the following types as specified:

Type I - Cylindrical Multi-perforated Grains
Type II - Cylindrical Single-perforated Grains

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-C-63346	-Charge, Propelling, 155MM, XM211, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-652	-Propellants, Solid, for Cannons, Requirements and Packing

FSC: 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (see 6.3)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9331701	-Charge, Propelling, XM211 (Green Bag) for 155mm Howitzer M198, M109A2/A3.

(Copies of specifications, standards, drawings and publictions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMC PAMPHLET 706-114

Experimental Statistics, Section 5, Tables

(Applications for copies should be addressed to the Publications and Reproduction Agency, Letterkenny Army Depot, Chambersburg, PA 17201.

3. REQUIREMENTS

3.1 Materials. The propellant shall be M1 with 1.0 ± 0.3% potassium sulfate added, Type I for zones 4 through 6 and the combination of Type II and Type I, potassium sulfate added for zone 3 conforming to the requirements of MIL-STD-652 as determined in 4.4.2.1.

3.2 Ballistic assessment requriements. The propellant when fired in the M199 Cannon with the projectile (M549 (see 4.5.1)) shall comply with the following requirements at 70 degrees Fahrenheit (^{O}F) (21.1 Degrees Centigrade (^{O}C) (see 4.5.1)) and standard weapon conditions. The propellant shall be capable of assessment to a velocity and pressures specified in Table 1.

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Table 1

Velocity and Pressure			Pressure (psi)	
Zone	Proj	Velocity (fps)	Minimum	Maximum
3	M549	975	-	-
4	M549	1160	-	-
5	M549	1510	-	-
6	M549	1980	-	35,500

3.3 Ballistic uniformity requirements. The propellant when fired in the M199 Cannon with the projectile (M549) (see 4.5.1) shall comply with the following requirements when tested as specified in 4.5.1.2. All charges (reference, master, calibration and test) shall be manufactured in accordance with MIL-C-63346.

3.3.1 Uniformity of velocity. The standard deviation of velocity for the lot of zone 3 and zone 6 at 70°F shall not exceed 4.0 fps and 6.5 fps, respectively.

3.3.2 Average pressure

3.3.2.1 70°F. At 70°F and standard weapon conditions, the propellant shall not produce an average pressure greater than 35,500 psi.

3.3.2.2 145°F. At 145°F and standard weapon conditions, the average pressure plus three (3) standard deviation shall not exceed the IMP (see 3.3.3).

3.3.3 Individual maximum pressure (IMP). The propellant shall not produce a greater than 38,000 psi at 70°F and not greater than 40,500 psi at 145°F when fired in a standard weapon.

3.4 Initial production. This specification contains provisions for initial production inspection. Requirements for the submission of initial production inspection samples by the contractor shall be as specified in 4.5.1.2.1.

3.5 Preproduction inspection. This specification contains provisions for preproduction inspection. Requirements for the submission of preproduction inspection samples by the contractor shall be as specified in 4.3.1.

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3.6 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspection shall be conducted on the unit product:

a. Preproduction Sample Inspection
b. Quality Conformance Inspection

4.3 Preproduction sample (Pilot lots). The preproduction sample is intended to provide the necessary information to establish the propellant granulation needed to meet the required ballistic parameters.

4.3.1 Submission. The Contractor shall select three (3) propellant granulations which, from experience, should best fit the requirements of this specification. The contractor shall submit a preproduction sample as designated by the Contracting Officer for evaluation in accordance with 4.3.2, consisting of three (3) pilot lots of approx. 1000 lbs each representing the three (3) candidate granulations.

4.3.2 Inspections to be performed. Each of the pilot lots comprising the preprodution sample and submitted in accordance with 4.3.1 may be subjected by the Government to any or all of the requirements of this specification.

4.3.3 Rejection. Government Engineering and Product Assurance personnel and the Contractor shall review the pilot lot test data and shall select a granulation size which best fits the requirements of this specification. The Technical Agency shall

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notify the Contracting Officer of the selected granulation. The contractor shall request approval from the Contracting Officer to produce the selected granulation. If none of the pilot lots submitted meets the requirments of this specification, a new preproduction sample shall be submitted in accordance with 4.3.1. All future production lots should be of the same basic granulation.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination and testing.

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.2.1 Sampling for laboratory testing (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical/physical properties in accordance with MIL-STD-652. (see 6.4 and 6.5).

4.4.2.2 Sampling for ballistic testing (see 3.2). Each lot of propellant shall be sampled for Proving Ground tests in accordance with MIL-STD-652.

4.4.2.3 Charge establishment (see 3.2). If the lot does not meet the applicable requirements, testing shall cease and the Test Director shall immediately notify the procuring activity.

4.4.2.4 Velocity uniformity (70°F) (see 3.3.1). If the product of the factor (see 6.7) times the velocity standard deviation of the rounds tested in accordance with 4.5.1.2 exceeds the applicable requirement, apply referral criteria in accordance with 4.5.1.5. If the referral criteria do not apply, the lot shall be rejected.

4.4.2.5 Average pressure (see 3.3.2). If the average pressure obtained exceeds the applicable requirement, apply referral criteria in accordance with 4.5.1.5. If the referral criteria cited do not apply, the lot shall be rejected.

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4.4.2.6 Individual maximum pressure (IMP) (see 3.3.3). If any individual pressure observed during the uniformity series, when corrected to standard conditions and recommended charge weight, exceeds the applicable requirement, the lot shall be rejected.

4.5 Test methods and procedures

4.5.1 Proving ground assessments (see Table II) This test shall be performed at a Government Proving Ground in accordance with the applicable acceptance test procedure using M549 projectiles inert loaded within 0.1 lb of an as fired weight of 96.0 lb using Primer, Percussion, M82 and fired in a M199 Cannon with 95 percent (minimum) remaining life (see 6.8). A spindle to charge standoff of 1.0 in ± 0.25 in shall be used. All charges shall be temperature conditioned in a large conditioning box, with adequate air circulation to assure uniform conditioning, at the required temperature for a minimum of 24 hours and assurance will be made that, when fired, the charges are at the conditioning temperature. The following tolerances shall apply to all temperature conditionings: -65°F + 5°F; 70°F ± 2.5°F; 145°F, -5°F. Copper crusher gages (2) shall be used to measure the maximum chamber pressure of each charge. Velcity coils will be used to measure the muzzle veloctiy of each charge. Test charges shall be assembled in accordance with drawing 9331701 and MIL-C-63346.

4.5.1.1 Charge establishment

4.5.1.1.1 Charge slope method. Three rounds at each zone loaded with a selected charge weight of propellant (weight selected should bracket expected service charge weight) shall be conditioned and fired at 70°F (21.1°C). Three calibration rounds shall also be conditioned and fired at 70°F (21.1°C), prior to the test rounds. From the data obtained, a charge weight shall be calculated which will yield the prescribed service velocity (see 3.2). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.5.1.2.1). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established. The resultant will be the recommended charge weight.

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TABLE II

PROVING GROUND TEST SUMMARY

Test Phase	Zone	Temperature	Sample size		Velocity Requirements		Pressure Requirements		
					Service Vel	Std Dev	IMP	Avg Press	Differential Pressure
		(°F)	Cal	Test	(fps)	(fps)	(psi)	(psi)	(psi)
Charge	3	70	3	3	975	N/A	N/A	-	N/A
	4	70	3	3	1160	N/A	N/A	-	N/A
	5	70	3	3	1510	N/A	N/A	-	N/A
Establishment	6	70	3	3	1980	N/A	N/A	35500	N/A
Uniformity	6	-65*	N/A	10	N/A	N/A		N/A	N/A
	3	70	10	10	975	4.0	-	-	N/A
	6	70	10	10	1980	6.5	38000	35500	N/A
	6	145	N/A	10	N/A	N/A	40500	**	Info

*Performed for initial production testing only

**The average pressure plus three (3) standard deviation shall not exceed the IMP.

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4.5.1.1.2 Constant slope method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite charge weight versus velocity versus pressure curve can be established from test results (see 4.5.1.1.1). This curve can then be used as in 4.5.1.1.1 to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method (see 4.5.1.1.1). The resultant will be the recommended charge weight.

4.5.1.2 Uniformity

4.5.1.2.1 Initial production. Thirty (30) charges shall be assembled with the calculated charge weight (see 4.5.1.1.1) of propellant at zone 6 and ten (10) charges at the calculated charge weight (see 4.5.1.1.1) at zone 3. Ten zone 6 charges each shall be conditioned to -65°F, 70°F and 145°F and ten zone 3 charges at 70°F (as indicated by thermocoupled control charge). Charges fired at 70°F shall be fired alternately with calibration charges and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditons and standard conditons). Firing results at all temperatures shall be corrected to standard projectile weight (see 4.5.1) and to recommended charge weight conditions (i.e., the difference between calculated (as fired) charge weight and recommended charge weight). The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. The initial production lots represent the first ten consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification.

4.5.1.2.2 Subsequent production. Twenty (20) rounds shall be assembled with the calculated charge weight (see 4.5.1.1.2) of propellant at zone 6 and ten (10) rounds at the calculated charge weight (see 4.5.1.1.2) at zone 3. Ten zone 6 rounds each shall be conditioned to 70°F and 145°F and ten zone 3 rounds at 70°F (as indicated by thermocoupled control round). Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions).

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Firing results at both temperatures shall be corrected to standard projectile weight (see 4.5.1) and to recommended charge weight conditions (i.e., the difference between as fired charge weight and recommended charge weight). The means and standard deviations of the velocities and pressures at all firing tempratures shall be calculated using corrected results. Throughout production, if 3 consecutive lots fail on first test, testing shall revert back to the inititial production plan (see 4.5.1.2.1).

4.5.1.3 Differential pressure and ignition delay (informational). Two pressure-time gages at different locations (one in the spindle and one at the projectile base) shall be used to record pressure-time traces on all +145°F firings. Differential pressure traces shall be obtained by subtracting the projectile gage output from the spindle gage output. The differential pressure of any charge is defined as the maximum amplitude of the first negative wave of the differential pressure trace prior to peak pressure. Any trace which contains no amplitude in the negative direction is considered to have a maximum differential pressure of zero. Ignition delay, the time from primer initiation to a rise in pressure equivalent to ten percent of the maximum expected pressure, shall be recorded on all differential pressure firings.

4.5.1.4 Retest criteria. A lot shall be retested whenever any of the following conditions exist or whenever the proving ground deems it necessary.

a. The final adjustment in charge weight exceeds 5.0 ozs of the charge weight used in uniformity firings, retest for uniformity in accordance with 4.5.1.2.

b. If for any reason, the Proving Ground considers that the conditions have detrimentally affected the test results, additional charges as required shall be tested.

4.5.1.5 Referral criteria. A test lot shall be referred to the procuring activity for disposition whenever any of the following conditions exist or whenever the proving ground deems it necessary.

a. Calibration rounds fired in any test phase fail to comply with Section 3 of this specification.

b. Calibration correction applied exceeds 2 percent of expected muzzle velocity.

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c. Calibration components are not available.

d. Number of test or calibration results utilized in calculations is less than specified.

e. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, stickers, poor projectile flights, etc. Each such event shall be reported in detail on the applicable firing record.

f. Test lot fails velocity standard deviation or average pressure requirement at 70°F and any extreme test value is an outlier. The following test shall be perfromed to determine if a single test result is an outlier. With a sample size of 10 at a significance level of 0.05, the critical value (T_C) is 0.477. Calculate a test vlaue (T_t) as follows:

For Low outlier

$$T_t = \frac{(X_2 - X_1)}{(X_9 - X_1)}$$

For High Outlier

$$T_t = \frac{(X_{10} - X_9)}{(X_{10} - X_2)}$$

Where:

X_1 = test result being tested as low outlier
X_2 = test result adjacent to X_1, after all the test results are ordered from smallest to largest
X_9 = Test result adjacent to the largest test result
X_{10} = test result being tested as high outlier

X_1 or X_{10} is an outlier if T_t is greater than T_C.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652, except that all propellant designated for use as reference shall be packed level A (see 6.5).

6. NOTES

6.1 Ordering data. Ordering data shall be in accordance with MIL-A-48078 with the following exceptions:

6.1.1 Procurement requirements. Procurement requirements of MIL-A-48078 shall apply.

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6.1.2 Contract data requirements. One copy each of the Propellant Description Sheet and the Acceptance Test Summary shall be forwarded to each office designated by the contracting officer. Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity.

6.2 Proving ground test summary. (see Table II)

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Devleopment Command (ARRADCOM) may also include drawings prepared by, and identifed as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.4 Surveillance sample. Samples shall be forwarded to US Army Armament Research and Development Command, Dover, NJ 07801, ATTN: DRDAR-LCE-P for the propellant surveillance test program in accordance with MIL-STD-652.

6.5 Reference sample. Whenever a propellant lot is selected for reference use, a 5-lb (2.3 kg) sample shall be forwarded to US Army Armament Research and Devlopment Command, Dover, NJ 07801, ATTN: DRDAR-LCE-P for use in evaluating the propellant stockpile.

6.6 Propellant Assessment. Propellant assessment may be required for a specifiec production lot of projectiles whose average as fired weight has been found to significantly deviate from the standard projectile weight for reasons other than projectiles design modifications.

6.7 Standard deviation. The factor, as taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limits (A.05) and n-1 degrees of freedom, makes allowance for the probability that a standard deviation from a smaller sample size exceeds the true standard deviation by chance alone. The standard deviation shall be calculated with (n-1) as the divisor in a standard statistical technique equivalent to that shown in MIL-A-48078.

6.8 Type of tube. Tubes used for these assessments shall be restricted to XM211, BAT's only.

Custodian:
Army-AR

Preparing Activity

Project Number: 1376-A162

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